# How to handle LLMs in your research

Pablo Mosteiro, Robert Bagheri, Özgür Togay, Laurence Frank
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Department of Methodology & Statistics

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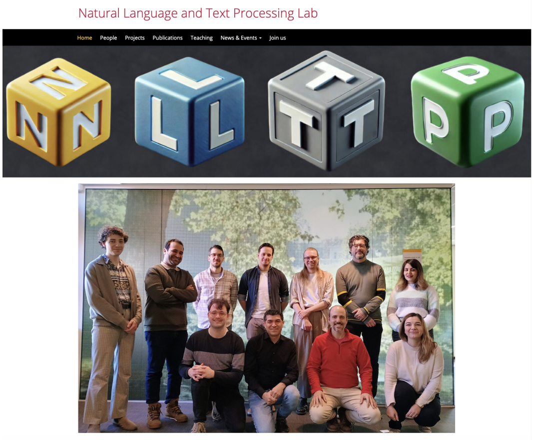
Universiteit Utrecht

# Today's Agenda

- Introduction (3 min)
- How LLMs Work (10 min)
- Limitations and Risks (10 min)
- Prompting CAs and Chatbots (10 min)
- Practical LLM Use in Research (10 min)
- Conclusion (2 min)

# About us




`https://nlp.sites.uu.nl/`

# Introduction: What do you know about LLMs?

- **Activity: Think-Pair-Share** (2 min)

Image produced in DALL-E December 2023 (Credit: Christine Fox)

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# Artificial Intelligence, ML, and LLMs: Key Definitions

**Artificial Intelligence (AI)**

- Computer systems performing tasks requiring human intelligence
- Examples: Chess playing, image recognition, language translation

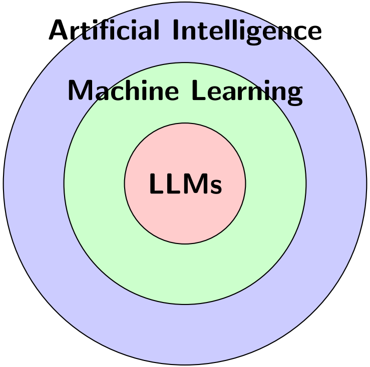
**Machine Learning (ML)**

- Subset of AI using algorithms that improve through experience
- Learns patterns from data without explicit programming

**Large Language Models (LLMs)**

- AI systems trained on vast text amounts
- Understand and generate human language
- Examples: GPT, Claude, Qwen, Mistral

# AI vs ML vs LLM: The Relationship




LLMs are a specific type of ML, which is a subset of AI

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# Model Learning Process: Training Phase

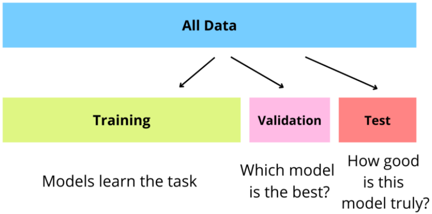
**Training Phase**

- Model learns patterns from large datasets
- Adjusts model to minimize prediction errors
- Requires massive computational resources
- Can take weeks or months for large models

**Example**: LLM learns that "cat" often appears with "dog" in pet-related contexts

# Model Learning Process: Validation Phase

## Validation and Testing




Source: `https://dqlab.id/5-panduan-praktis-membuat-model-machine-learning`

**Key insight**: Good models **generalize**, not just **memorize**

# Prompting phase

U weet wat ik ga zeggen

Credit for this slide: Albert Gatt & Dong Nguyen

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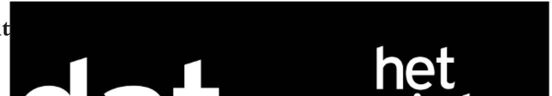
# Prompting phase

U weet wat ik ga

# Model Learning Process: Biases

**Common Biases in LLMs**

- **Training data bias:** Reflects biases in source material
- **Representation bias:** Underrepresents certain groups
- **Confirmation bias:** Reinforces existing beliefs
- **Temporal bias:** Becomes outdated as language/society changes

**Real-world impact**: Biased models can perpetuate harmful stereotypes

# What is Generative AI?

**Definition**: AI that creates new content (text, images, code, audio)

**Key Features**:

- Creates novel outputs, not just classifies
- Learns patterns from training data
- Generates human-like content

**Examples**: ChatGPT (text), DALL-E (images), CodeT5 (code)

# Text-GenAI components

| consists of | fed by | GPT learns |
|---|---|---|
| **LLM** | textual internet data | "knowledge" of language and content |
| **Dialogue Model** | social media, forums | "knowledge" of dialogue structure |
| **Task Model** | programmers | tasks, self knowledge and limits |
| **Ethical Governor** | Kenian data labellers | ethical answers |

**Reinforcement Learning**

Credit for this slide: Davitze Könning

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# Generative AI vs. Other Models

| Feature | Generative | Discriminative |
|---|---|---|
| Primary Task | ? | ? |
| Output Type | ? | ? |
| Example | ? | ? |

# Generative AI vs. Other Models

| Feature | Generative | Discriminative |
|---|---|---|
| Primary Task | Create new content | Classify existing data |
| Output Type | Novel text/images | Labels/categories |
| Example | GPT, DALL-E | Spam filters, medical diagnosis |

# Activity: True or False?

**Instructions**: Raise your hand if you think the statement is TRUE

**Statement 1**: Generative AI can only work with text

**Statement 2**: All generative models are also discriminative

**Statement 3**: Generative AI creates completely original content never seen in training

**Statement 4**: LLMs are a type of generative AI

# Limitations and Risks: Hallucinations

**What are hallucinations?**

- LLMs generate false or nonsensical information
- Appears confident and plausible
- Can include fake citations, non-existent facts

# Limitations and Risks: Bias

## Types of Bias

- **Gender bias:** Assumes doctors are male, nurses are female
- **Racial bias:** Associates certain groups with negative stereotypes
- **Cultural bias:** Primarily represents Western perspectives
- **Academic bias:** Favors published, English-language sources

# Limitations and Risks: Output Reliability

**Reliability Issues**

- **Inconsistency:** Same prompt can yield different answers
- **Context dependency:** Performance varies by domain
- **Temporal drift:** Knowledge becomes outdated
- **Prompt sensitivity:** Small changes affect outputs significantly

**Best practice**: Always verify critical information from multiple sources

# Limitations and Risks: Ethics

## Ethical Concerns

- **Data:** Training data may include personal or copyrighted material
- **Environmental impact:** High energy consumption
- **Labor practices:** Data annotation often involves poor working conditions

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  "When it gets to the day before payday, they close the account and say that you violated a policy," Kanyugi said.

  Employees say they have no recourse or even a way to complain.

  The company told 60 Minutes that any work done "in line with our community guidelines was paid out." In March, as workers started complaining publicly, Remotasks abruptly shut down in Kenya, locking all workers out of their accounts.

  **The mental toll of AI training**

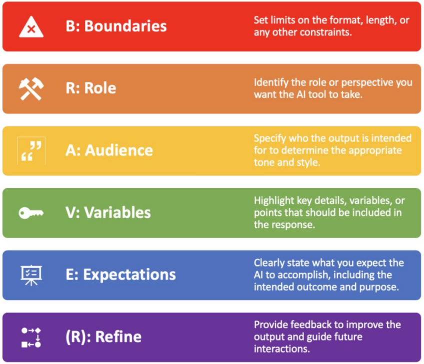
  Workers say some of the projects for Meta and OpenAI also caused them mental harm. Wambalo was assigned to train AI to recognize and weed out pornography, hate speech and excessive violence from social media. He had to sift through the worst of the worst content online for hours on end.

  "I looked at people being slaughtered," Wambalo said. "People engaging in sexual activity with animals. People abusing children physically, sexually. People committing suicide."

  Berhane Gebrekidan thought she'd been hired for a translation job, but she said what she ended up doing was reviewing content featuring dismembered bodies and drone attack victims.

  `https://www.cbsnews.com/news/ai-work-kenya-exploitation-60-minutes` (Nov 2024)
- **Academic integrity:** attribution of AI assistance / uploading restricted data
- **Open source:** Often little documentation on data, training, architecture

# Effective Prompting: BRAVE(R) Framework




*Prompt formulation by Christine Fox, in collaboration with ChatGPT (May 2024).*

**Poor prompt**: "Summarize this paper"

**Better prompt**: "Act as a psychology research assistant [R]. Summarize this paper in 3 bullet points [B] focusing on methodology and findings relevant to cognitive behavioral therapy [V]. Use academic language [A] and APA style [E]."

**Key insight**: Specific, contextual prompts yield better results

# Evaluating and Critically Assessing GenAI Responses

Prompt formulation by Christine Fox, in collaboration with ChatGPT (May 2024).

**F: Focus** — Focus on the AI's response by breaking it down into its main points and arguments.

**A: Authenticate** — Authenticate the AI's output by verifying it against current research and empirical data.

**C: Critique** — Critique the response's accuracy, relevance, and depth in the context of the topic.

**T: Think** — Think about the response from various perspectives and its potential impact.

**S: Scrutinise** — Scrutinise the AI's conclusions by critically questioning and exploring alternative hypotheses or interpretations.

More info:

`https://teachersguidegsls.nl/genai/generative-ai-guidelines/`

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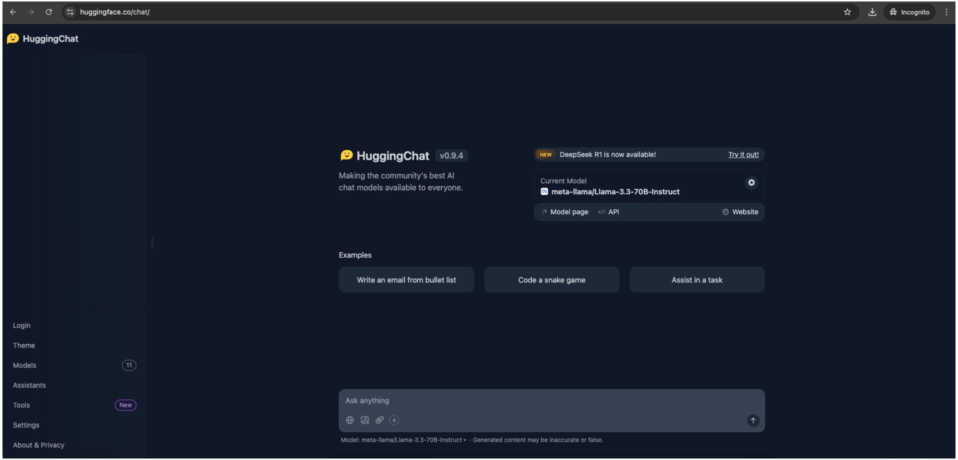
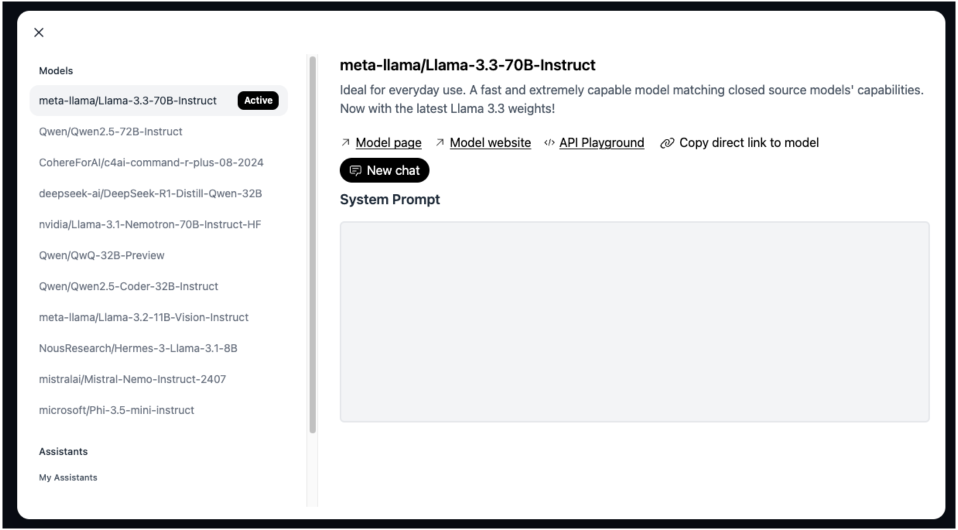
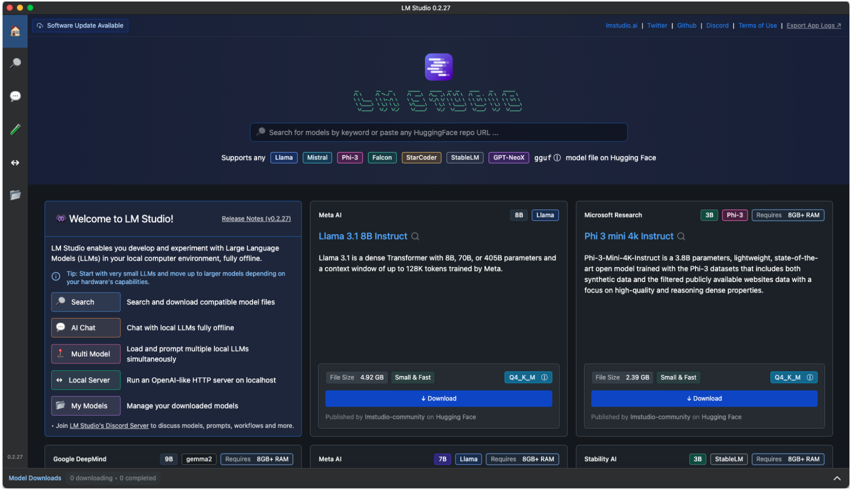
# Ways to prompt



# Switch models

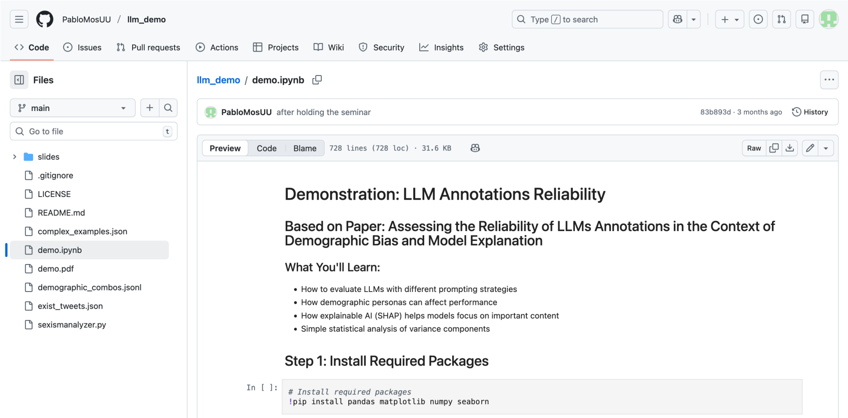


# Using an LLM locally (no server)



# Versatile platform

- OpenRouter.ai




https://github.com/PabloMosUU/llm_demo

# Official UU policy

`https://www.uu.nl/en/organisation/ai-policy/researchers`

- Benefits and use cases
- Risks and limitations
- Effective use of GenAI (*Background check*)
- Citing GenAI in your work

# LLM Use in Research

**Appropriate & Inappropriate Uses**

- Think-Pair-Share

# LLM Use in Research: Appropriate Applications

## Appropriate Uses

- **Literature review:** Summarizing papers (with verification)
- **Writing assistance:** Drafting, editing, proofreading
- **Code generation:** Writing analysis scripts

> **Caveat**
>
> **only** when you know which statistical analysis is suitable

- **Brainstorming:** Generating research ideas
- **Translation:** Working with multilingual sources

**Key**: Always maintain human oversight and verification

# LLM Use in Research: Inappropriate Applications

**Avoid Using LLMs For:**

- **Statistical analysis:** LLMs can't perform proper statistical tests
- **Fact-checking:** They may hallucinate information
- **Critical decisions:** Participant safety, ethical approvals
- **Original thinking:** Generating truly novel theories
- **Replacing expertise:** They supplement but don't replace researcher knowledge

# Working with others

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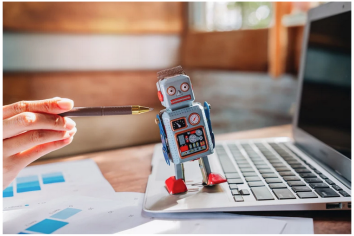
## Don't let your students use AI as a ghostwriter

**How a research proposal generated by artificial intelligence transformed my approach to teaching and supervision.**

By Yanjun Shen



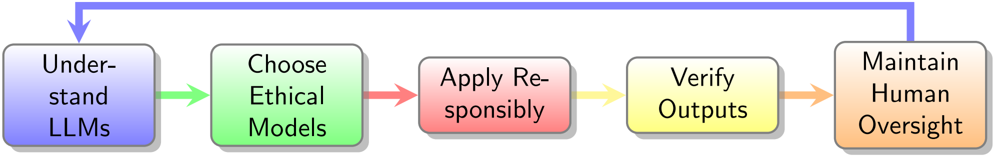
Using generative artificial intelligence models to kick ideas around is more effective than outsourcing thinking wholesale, says Yanjun Shen. Credit: iabkitticha/iStock via Getty

1. thinking for yourself
2. read literature
3. ask AI for advice
4. critically analyze

# What we've done

- How LLMs Work
- Limitations and Risks
- Prompting CAs and Chatbots
- Practical LLM Use in Research

# Key Takeaways: LLM Workflow




**Remember:**

- LLMs are powerful tools, not replacements for human expertise
- Always prioritize ethics, privacy, and transparency
- Verify outputs and maintain critical thinking

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